

Core Output Control Parameters

These parameters control **how the model selects the next token during generation**. They operate at the probability distribution level.

1 Temperature — Controls Randomness

Range: 0.0 – 2.0

(Default is usually around 0.7–1.0 depending on model)

What It Does

Temperature scales the probability distribution before token selection.

- **Low temperature (0–0.3)**
 - More deterministic
 - Safer
 - More repetitive
 - Good for factual answers, summaries, code
- **Medium (0.5–0.8)**
 - Balanced
 - Natural sounding
 - Default for most tasks
- **High (1.0–2.0)**
 - More creative
 - Riskier

- Higher variability
- More hallucination potential

Technical Insight

Lower temperature sharpens the distribution.
Higher temperature flattens it.

It does not change knowledge — it changes selection randomness.

2 Top-p (Nucleus Sampling)

Range: 0.0 – 1.0

What It Does

Instead of sampling from all tokens, the model selects from the smallest group of tokens whose cumulative probability $\geq p$.

Example:

- `top_p = 0.9`
→ Only the top 90% probability mass is considered.

Effect

- Lower values → safer, more predictable
- Higher values → more diverse

Temperature vs Top-p

- **Temperature** reshapes probability distribution
- **Top-p** truncates the distribution

They are often used together.

Best practice: adjust one primarily, not both aggressively.

3 Frequency Penalty

Range: typically 0.0 – 2.0

What It Does

Penalizes tokens based on how often they have already appeared in the response.

Effect

- Reduces repetition
- Encourages lexical diversity
- Good for long outputs

High values may cause awkward wording.

4 Presence Penalty

Range: typically 0.0 – 2.0

What It Does

Penalizes tokens that have appeared at least once.

Difference from frequency penalty:

- **Frequency penalty** scales with repetition count
- **Presence penalty** applies once per token regardless of frequency

Effect

Encourages topic shifts and introduction of new ideas.

Related & Advanced Concepts

To deepen this section, consider adding the following:

5 Max Tokens

Hard cap on output length.

- Controls cost
 - Prevents runaway generation
 - Only guaranteed limiter
-

6 Stop Sequences

Predefined strings that terminate generation.

Example:

```
stop=[ "END" ]
```

Used in structured outputs and pipelines.

7 Logit Bias

Manually increases or decreases probability of specific tokens.

Use cases:

- Force structured formatting

- Prevent unwanted words
- Enforce brand compliance

This is fine-grained probability manipulation.

8 Deterministic vs Stochastic Decoding

- **Deterministic** → temperature ≈ 0
- **Stochastic** → temperature > 0

Deterministic outputs are reproducible.
Stochastic outputs vary per run.

9 Greedy Decoding

Always selects highest-probability token.

Equivalent to temperature = 0 behavior.

Very stable. Low creativity.

10 Beam Search (Advanced Systems)

Explores multiple token paths simultaneously and selects highest scoring sequence.

More computationally expensive.
Often used in translation tasks.

How These Parameters Work Together

Parameter	Controls	Best For
Temperature	Randomness	Creativity vs precision
Top-p	Probability cutoff	Diversity control
Frequency penalty	Repetition level	Long outputs
Presence penalty	Topic novelty	Brainstorming
Max tokens	Length	Cost control
Stop sequences	Hard stopping	Structured output

Practical Configuration Patterns

Factual / Technical Answer

- Temperature: 0.2
 - Top-p: 1.0
 - Frequency penalty: 0
 - Presence penalty: 0
-

Creative Writing

- Temperature: 0.9–1.2
 - Top-p: 0.9
 - Frequency penalty: 0.5
 - Presence penalty: 0.6
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Brainstorming Ideas

- Temperature: 1.0
 - Top-p: 0.95
 - Presence penalty: 0.8
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Key Strategic Insight

These parameters do not change what the model knows.
They change:

- Probability shaping
- Token selection behavior
- Diversity vs determinism tradeoff

They are **inference-time control levers**, not training-time modifiers.

Where You Can Access These Attributes

These parameters are **not visible in normal ChatGPT conversation mode**.

They are available when you interact with LLMs through:

1 Official APIs

- OpenAI API
- Azure OpenAI
- Anthropic API
- Google Gemini API

You configure them inside the request payload (e.g., `temperature`, `top_p`, `max_tokens`).

2 Developer Platforms

- Playground interfaces
- API dashboards
- Model testing consoles

These platforms expose sliders or input fields for these controls.

3 Open-Source Model Runtimes

- Local LLM frameworks (e.g., inference servers)
- Hugging Face model pipelines
- Custom backend implementations

Most open-source inference libraries expose these parameters directly.

Important Clarification

In consumer chat interfaces:

- You cannot directly set `temperature`, `top_p`, penalties, etc.
- The provider preconfigures them internally.

To control these attributes precisely, you must use an **API or developer environment**.
